

Clinical Trial Executive Report

Drug X vs. Placebo — Phase III Efficacy & Safety Analysis

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1. Trial Overview & Demographics

Metric	Value
Total patients enrolled	2000
Drug X arm	1000
Placebo arm	1000
Mean age (range)	54.8 (30-75)
Female patients	54.6%
Trial completion rate	86.4%
Dropout rate	13.6%

2. Primary & Secondary Endpoint Results

On average, the Drug X arm scored 9.05 points higher than Placebo on primary outcome score (0-100) (60.19 vs 51.14). This difference is statistically significant (Welch's $t=18.05$, $p<0.001$), meaning we would see a difference this large by chance in fewer than 0.1% of repeated trials. The effect size is large (Cohen's $d=0.81$).

A chi-square test of independence was run on HbA1c <7% control at Week 24 across treatment arms ($\chi^2=218.75$, $\text{dof}=1$, $p<0.001$). There is a statistically significant association between treatment arm and HbA1c <7% control at Week 24.

3. Safety Profile

A chi-square test of independence was run on adverse event occurrence across treatment arms ($\chi^2=127.61$, $\text{dof}=1$, $p<0.001$). There is a statistically significant association between treatment arm and adverse event occurrence.

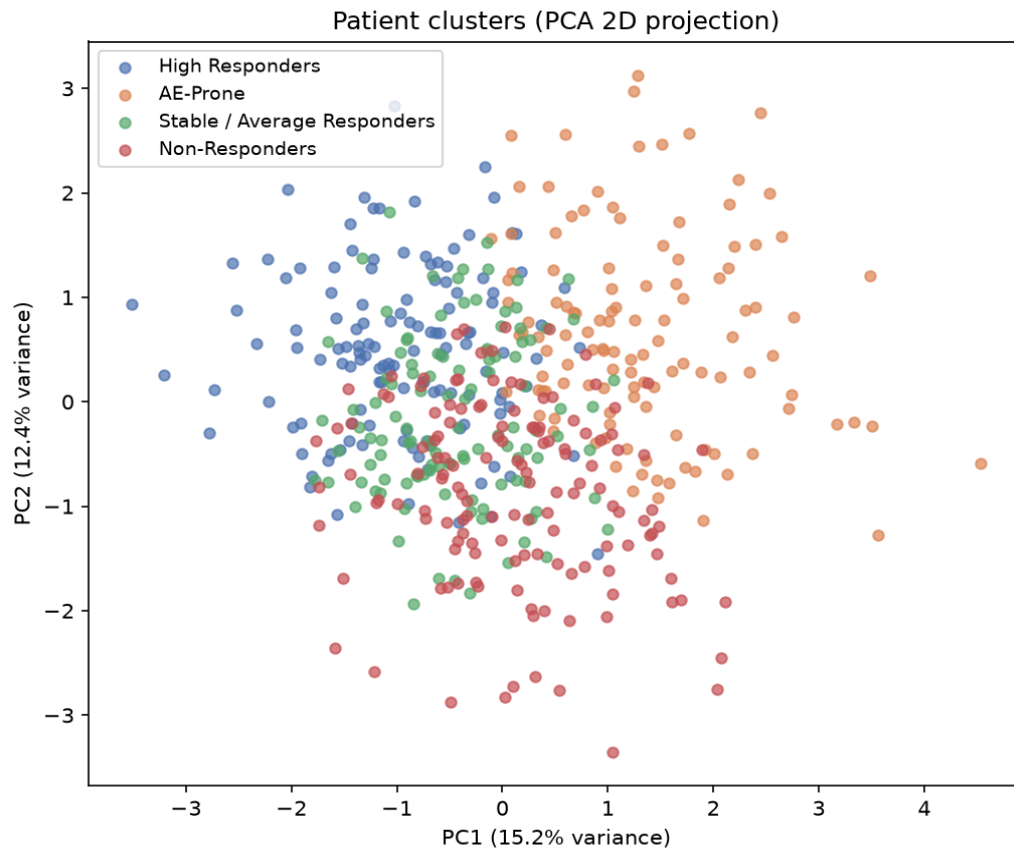
A chi-square test of independence was run on adverse event severity class across treatment arms ($\chi^2=196.71$, $\text{dof}=2$, $p<0.001$). There is a statistically significant association between treatment arm and adverse event severity class.

A Kruskal-Wallis test compared HbA1c reduction by Week 24, grouped by worst AE severity across 3 groups without assuming normally distributed data ($H=83.74$, $p<0.001$). The groups differ significantly in HbA1c reduction by Week 24, grouped by worst AE severity. Median values: No AE=0.40, Mild AE=0.67, Severe AE=0.58.

4. Patient Cluster Profiles (K-Means Segmentation)

K-Means with k=3 clusters (silhouette score 0.082).

Cluster	N	Label	Outcome Score	Any-AE Rate	Severe-AE Rate	Dropout Rate
0	748	Non-Responders	57.438	47.9%	0.1%	8.4%
1	772	High Responders	59.963	50.5%	0.0%	8.5%
2	480	AE-Prone	45.984	99.8%	33.5%	29.8%

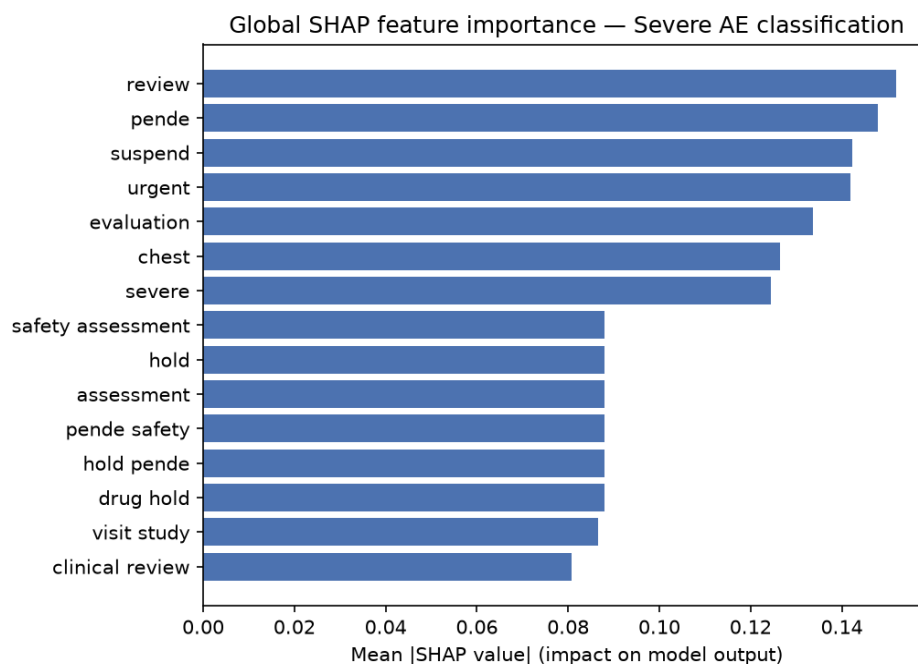


5. NLP Adverse-Event Classifier & Explainability

Metric	Value
Accuracy	92.7%
Precision (macro)	78.0%
Recall (macro)	89.9%
F1 (macro)	0.829

On the held-out test set, the NLP classifier detected 483 adverse-event notes vs. 255 caught by manual CRF coding (89.4% uplift), and recovered 96.9% of the 97 adverse events manual coding missed entirely.

SHAP global feature importance — Severe AE classification:



6. Top Clinical Insights

The following insights and recommendations are generated by Agent 3 (Insight Generation Agent), a live LLM call grounded in the statistics, clustering, and classifier tool results above.

1. The trial's primary endpoint shows a statistically significant difference between the Drug X and Placebo arms, with an average score 9.05 points higher for Drug X, and a large effect size of Cohen's $d = 0.81$.
2. The AE classifier achieves a high accuracy of 0.9271 and a weighted F1 score of 0.9323, with a business impact analysis showing an uplift of 89.4% in detected adverse events compared to manual coding.
3. Cluster analysis reveals three distinct patient profiles: "Non-Responders" with a mean outcome score of 57.438, "High Responders" with a mean outcome score of 59.963, and "AE-Prone" patients with a high severe AE rate of 0.335 and a mean outcome score of 45.984.
4. The chi-square test of independence shows a statistically significant association between treatment arm and adverse event occurrence, with a chi-square value of 127.61 and $p < 0.001$.
5. The search of clinical notes reveals that patients generally report feeling well, with some instances of mild adverse events such as feeling unsteady on their feet, but no severe adverse events are reported in the first 24 weeks.

7. Recommendations for Next Steps

1. Based on the statistically significant primary endpoint results, with a p-value of less than 0.001, it is recommended to proceed with further development of Drug X.
2. The high accuracy and weighted F1 score of the AE classifier, at 0.9271 and 0.9323 respectively, suggest that it can be reliably used to detect adverse events, and its implementation can lead to an 89.4% uplift in detected adverse events.
3. The distinct patient profiles identified through cluster analysis, with "Non-Responders" making up 37.4% of the patient population, suggest that personalized treatment approaches may be beneficial, with tailored strategies for each cluster to improve outcomes and reduce adverse events.